

- 6.11 understand that there is a force on a charged particle when it moves in a magnetic field as long as its motion is not parallel to the field**
- 6.12 understand that a force is exerted on a current-carrying wire in a magnetic field, and how this effect is applied in simple d.c. electric motors and loudspeakers
- 6.13 use the left hand rule to predict the direction of the resulting force when a wire carries a current perpendicular to a magnetic field
- 6.14 describe how the force on a current-carrying conductor in a magnetic field increases with the strength of the field and with the current.

d) Electromagnetic induction

Students will be assessed on their ability to:

- 6.15 understand that a voltage is induced in a conductor or a coil when it moves through a magnetic field or when a magnetic field changes through it and describe the factors which affect the size of the induced voltage
- 6.16 describe the generation of electricity by the rotation of a magnet within a coil of wire and of a coil of wire within a magnetic field and describe the factors which affect the size of the induced voltage
- 6.17 describe the structure of a transformer, and understand that a transformer changes the size of an alternating voltage by having different numbers of turns on the input and output sides**
- 6.18 explain the use of step-up and step-down transformers in the large-scale generation and transmission of electrical energy**
- 6.19 know and use the relationship between input (primary) and output (secondary) voltages and the turns ratio for a transformer:**

$$\frac{\text{input (primary) voltage}}{\text{output (secondary) voltage}} = \frac{\text{primary turns}}{\text{secondary turns}}$$

$$\frac{V_p}{V_s} = \frac{n_p}{n_s}$$